Neuryo v5.2 prototype buy plan

7 Nov 2026 · one order, ₹50,000 grant · prices read on the seller pages on 7 Nov unless marked est. (yellow rows) · USD at ₹94.05 mid-market (x-rates, 7 Nov 16:25 UTC), ₹97.34 on the card (about 3.5% markup)

What	When	Total
Domestic orders: Upside Down Labs, Robu, Robocraze, Amazon, ToolBuy, chemist (frame mock-up, electrodes, consumables, bench electronics)	Order now	₹13,401
FRI TDE-211 Ag/AgCl pod cores × 4 (import, through VIT)	After the headphone mock-up passes on 10 heads	₹7,460
ADS1299 rev B board × 2, assembled at JLCPCB (import, through VIT)	After the schematic is reviewed by your professor	₹25,723
Total, imports through VIT		**₹46,584**
If the lab already has PETG, a 0.01 g scale and a caliper (−₹1,468)		₹45,116
If both imports go as personal imports instead (42.08% duty)		₹55,970

Recommendation. Order the domestic lists now (₹13,401), run the mock-up test, then place the two imports through VIT. That lands at about ₹46,584, ₹3,416 under the grant. The board is 55% of the money and the only line that is still an estimate in parts; get the real JLCPCB quote the day the Gerbers are ready. If the board goes as a personal import, you are about ₹5,970 over; the fallback below (gold cups instead of FRI) claws back about ₹6,491.

1 · Order now: domestic

Grouped by shop so each order clears its free-delivery limit where there is one.

Upside Down Labs

Item	For	Qty	Price	Line	Link
UDL colour snap cables, pack of 10	Leads for 9 head sites + 2 ECG; cut the plug end and solder to the board	2	₹1,999	₹3,998	store.upsidedownlabs.tech/product/different-color-s…
UDL Neo Gel electrodes, 100 pcs (3.2 × 2.5 cm)	Forehead strip pads (MVP), M1, M2, eye pad, ECG	1	₹799	₹799	store.upsidedownlabs.tech/product/neo-gel-electrode…
Upside Down Labs total	Free delivery above ₹999 (store page).			**₹4,797**	

Robu

Item	For	Qty	Price	Line	Link
Seeed Studio XIAO ESP32S3	One for bench firmware now, one soldered onto the board (replaces the ESP32-S3 DevKitC)	2	₹1,139	₹2,278	robu.in/product/seeed-studio-xiao…
Numakers PETG-HS 1.75 mm, 1 kg (skip if you have)	Pods, spine, carriages, temple arms, rear band, ear clip	1	₹780	₹780	robu.in/product/numaker-petg-hs-1…
Robu total	Free shipping above ₹499 (per GrabOn; confirm at checkout).			**₹3,058**	

Robocraze

Item	For	Qty	Price	Line	Link
MAX30102 pulse sensor module	Earlobe clip + test-kit forehead window	2	₹249	₹498	robocraze.com/products/max30102-pulse-o…
Witty Fox 3.7 V 1000 mAh Li-ion, with protection	Battery in the right pod; check pouch size on arrival	1	₹205	₹205	robocraze.com/products/3-7v-1000mah-li-…
8CH USB logic analyzer	Debug the ADS1299 SPI link	1	₹614	₹614	robocraze.com/products/8ch-usb-logic-an…
Robocraze total	Free delivery on orders of ₹999 and above (shipping policy page); this order is ₹1,317.			**₹1,317**	

Amazon.in

Item	For	Qty	Price	Line	Link
Koss KPH7 on-ear headphones	Spring arch + sliders for the mock-up (listing: metal and plastic)	1	₹1,000	₹1,000	amazon.in/dp/B006T9ZKAQ
Stainless compression spring assortment est.	v5.2 pod springs: pick three near 0.13 N/mm (1 N at about 7.5 mm) on the scale; price not read on a seller page	1	₹300	₹300	
OLEIO M3 × 12 mm screws with nuts	Arch sliders, pod shells, ear clip hinge	1	₹289	₹289	amazon.in/dp/B0HKJ7692J
Silicone rubber sheet 2 mm, 1 × 1 ft	Temple pads, rear hook pad, pod linings (replaces TPU filament)	1	₹459	₹459	amazon.in/dp/B0D243VWDB
Worison black foam sheets, A4 × 10	Light-blocking pad on the ear clip	1	₹175	₹175	amazon.in/dp/B09G3BDXYL
Araldite Standard 90 g	Seal pod lids, strain relief on soldered leads	1	₹195	₹195	amazon.in/dp/B01HS00QX0
Romsons alcohol swabs, 100	Skin prep	1	₹179	₹179	amazon.in/dp/B0G653LQPL
PVA sponge, pack of 2	Cut into foam tips for the pods	1	₹239	₹239	amazon.in/dp/B0BRGG1LFY
Flat 6-core 28 AWG cable, 5 m	Ear-clip lead (5 wires)	1	₹140	₹140	amazon.in/dp/B084V1TR88
ATOM pocket scale, 200 g × 0.01 g (skip if you have)	Pod force check (1 N ≈ 102 g) and spring picking	1	₹289	₹289	amazon.in/dp/B06XMZV7RH
IONIX 150 cm tailor tape	Head circumference, nasion–inion, size tape	1	₹99	₹99	amazon.in/dp/B098R1VVWX
Vinabo digital caliper 150 mm (skip if you have)	Print fit, pad and spring sizes	1	₹399	₹399	amazon.in/dp/B0BDLX8DWG
Amazon.in total	Delivery charges depend on seller; check at checkout.			**₹3,763**	

ToolBuy

Item	For	Qty	Price	Line	Link
Karam PN521 ratchet helmet	Cut out the rear knob + both toothed straps for the dial	1	₹306	₹306	toolbuy.com/tp/karam-isi-marked-shelm…
ToolBuy shipping est.		1	₹100	₹100	
ToolBuy total	Dispatch in 2 business days; shipping shown at checkout.			**₹406**	

Chemist

Item	For	Qty	Price	Line	Link
0.9% saline + three 1 ml syringes (no needle) est.	Wetting the pod tips	1	₹60	₹60	
Chemist total				**₹60**	

2 · Hold: imports (route both through VIT purchase)

Gate: the headphone mock-up passes the 10-head test (each pod 0.7–1.5 N on the scale, no sore spots).

FRI TDE-211 reusable Ag/AgCl snap electrode	Amount	Link
4 × $4.95 (3 pods + 1 spare; the 20-pack is sold out)	₹1,927	fri-fl-shop.com/products/reusable-snap-ag…
DHL shipping (from your 29 Oct cart; FRI calculates it at checkout) est.	₹4,400	
Duty and clearance through VIT (~10% + ₹500, 7 Nov plan assumption)	₹1,133	
FRI total through VIT (personal import: ₹9,490)	**₹7,460**	

Gate: rev B schematic and layout done and reviewed. Economic PCBA is fine because the XIAO is hand-soldered (no ESP32 module on the JLCPCB BOM). Leave the electrode input headers off the BOM: the leads are soldered in by hand.

ADS1299 rev B, 2 boards assembled (JLCPCB minimum)	USD	Link
4-layer PCB, 5 pcs, ~70 × 50 mm est.	$10.00	
Economic PCBA setup fee	$8.18	jlcpcb.com/solutions/pcb-prototype-a…
Stencil est.	$1.50	
SMT joints, 2 boards × ~450 × $0.0016	$1.44	jlcpcb.com/solutions/pcb-prototype-a…
Extended-part feeder fees, ~12 × $3 est.	$36.00	jlcpcb.com/help/article/98-pcb-assem…
ADS1299IPAGR × 2 at $70.817 (419 in stock)	$141.63	jlcpcb.com/partdetail/TexasInstrumen…
LM27762DSSR × 2 at $0.9617	$1.92	jlcpcb.com/parts/componentSearch
TPS7A2033PDBVR × 2 at $0.2082	$0.42	jlcpcb.com/parts/componentSearch
BMI270 × 2 at $2.5906	$5.18	jlcpcb.com/parts/componentSearch
TPD4E1B06DCKR × 10 at $0.4302	$4.30	jlcpcb.com/parts/componentSearch
Other passives, regulators, connectors est.	$10.00	
Shipping to India est.	$15.00	
Board total $235.57 × ₹97.34 = ₹22,930; through VIT (~10% + ₹500) **₹25,723**; personal import ₹33,079	**$235.57**	

3 · Cheaper options and what not to buy

Option	Effect	Link
UDL gold cup electrodes, pack of 10 (₹2,999, 10 mm, 1 m lead, 75 in stock) instead of FRI	Saves about ₹4,461 through VIT (₹6,491 vs a personal import), ships free in the UDL order. Trade-off: gold-plated, not Ag/AgCl, so more slow drift with saline; your semi-dry evidence is for Ag/AgCl. Use only if the board comes in over budget.	store.upsidedownlabs.tech/product/gold-cup-electrod…

Option	Effect	Link
Midline strap, webbing, elastic and steel strip: not needed	v5.2 holds Fz, Cz and Pz in spring carriages on a rigid PETG spine clipped to the arch at Cz, one soft spring per pod (from the assortment). Spine, saddle clip and carriages are PETG prints.	
TPU filament (₹2,149 in the old plan): drop it	The 2 mm silicone sheet (₹459) covers temple pads, hook pad and linings.	
ESP32-S3 DevKitC (₹1,299): drop it	The second XIAO does the bench firmware on the same chip and pins as the board.	
Also not needed	PA12 midline band (v4 only), brass heat-set insert kits (₹1,129+; use M3 nuts in printed pockets), torsion-spring kit (₹1,734; reuse a binder-clip or clothes-peg spring for the ear clip), separate BMI270 breakout (on the board), NuPrep and Micropore (alcohol swab prep only), PVA powder and glycerol for the hydrogel tip (only if the damp foam tip fails the drying test).	
Only if a test fails	Weaver Ten20 paste, 228 g, ₹1,890 on Amazon: only if Neo Gel on the mastoids fails the impedance check. Check the seal and expiry on arrival.	amazon.in/dp/B075C3M2M7

4 · Checks before paying

• Re-open each link on the day you pay; pay if the price is within 15% of this plan. Prices and stock change daily.
• Koss KPH7: the listing gives "Metal, Plastic". Before cutting, confirm the band is spring steel with sliders; if it is plastic, return it and use another headphone with a visible steel band.
• Witty Fox 1000 mAh: the page gives no size or connector. Check it fits the right pod and add a JST-PH 2.0 lead if it has bare wires.
• Pod springs: squeeze each on the pocket scale; keep three that give about 1 N (102 g) at 7.5 mm. Fitting: slide the arch sliders until the Cz fill tube band meets its port, and save the setting per patient.
• Ask VIT purchase whether the JLCPCB and FRI orders can go through the institute import route; the ~10% duty here is the 7 Nov plan's assumption, not a quote.
• Recheck ADS1299 stock and price at JLCPCB on the day you order ($70.817, 419 in stock on 7 Nov). The extended-part count (12) and PCB price are estimates until the BOM exists.
• Robu: the 1000–1500 mAh LiPo cells (WLY803040, WLY102050, WLY103048) were all out of stock on 7 Nov, which is why the battery is from Robocraze.

Sources opened 7 Nov 2026

Seller pages linked in each row. JLCPCB fees: PCB prototype assembly ($8.18 economic setup, $0.0016 per joint, 2-board minimum) and PCB assembly FAQs ($3 per extended part). Exchange rate: x-rates. Robocraze free delivery: shipping policy. Import duty, clearance and the FRI shipping figure carry over from your 7 Nov v4.4 buy plan.